Alexandria University
Faculty of Engineering
Computer and Communications program

Spring 2026
CSE 476 Data Mining
Assigned: 12th of March

Assignment 2

Non-linear Dimensionality Reduction

The objective of this assignment is to implement and evaluate different dimensionality reduction techniques for data visualization and classification. The techniques covered include Laplacian Eigenmaps, t-SNE, and UMAP.

Dataset Preparation

Use the following code snippet to generate the Swiss Roll dataset:

```
from sklearn.datasets import make_swiss_roll
import numpy as np

def generate_swiss_roll(n_samples=1500, noise=0.1, random_state=42):
   """Generate Swiss Roll dataset with binary labels."""
   X, color = make_swiss_roll(n_samples, noise=noise,
random_state=random_state)
   # Create binary labels based on color for classification task
   y = (color > np.median(color)).astype(int)
   return X, y, color
```

Exploratory Data Analysis (10)

Visualize the dataset using **2D and 3D scatter plots** to observe how the data points are arranged in a spiral-like shape. Use histograms or box plots to analyze the distribution of values. Discuss the main challenges of classifying data from the Swiss Roll dataset.

Laplacian Eigenmaps (20)

Implement Laplacian Eigenmaps from scratch, ***do not use external libraries***. Use different values of sigma: {0.5, 1, 2}. Use a different number of components {2,5,10,20}. Plot eigenvalues to show how they contribute to dimensionality reduction. Plot the **2D projection** of the dataset.

SNE and UMAP (30)

Use SNE and UMAP dimensionality reduction **(you don't need to implement this from scratch)**. You will have to use the same methods explained in the lecture and not just use default parameters. You will need to explain each parameter you have and try several values that you see suitable. Plot 2D scatter plots representing the most important features for each of the two cases.

Compare Computation Time (10)

Analyze the computation time of different Dimensionality reduction techniques under various conditions. Plot the bar plot showing the computation time of each dimensionality reduction technique.

Classification with SVM (20)

Train a **Support Vector Machine (SVM)** classifier on the reduced datasets. Compare classification accuracy using different embeddings. Run the SVM on the original dataset as well to compare. Use **Stratified K-Fold cross-validation** to measure generalization. Plot SVM decision boundaries in the reduced space. Plot a bar plot to compare with original data without dimensionality reduction

Report (10)

Prepare a comprehensive report detailing your methodology, results, and conclusions. Include visualizations, code snippets, and a discussion of the implications of your findings.

(Bonus) Neighborhood Preservation (20)

Compare the **neighborhood preservation rate** across different dimensionality reduction methods to assess their effectiveness in retaining local structure.

Neighborhood preservation rate measures how well a dimensionality reduction technique maintains the local structure of the data. Specifically, it checks how many of the nearest neighbors in the original high-dimensional space are still neighbors in the reduced low-dimensional space.

How It Works

1. ***Find k-Nearest Neighbors (k-NN) in High-Dimensional Space***
2. ***Find k-Nearest Neighbors in Low-Dimensional Space***
3. ***Compare Overlapping Neighbors***

4. ***Compute Preservation Rate or Jaccard similarity for better measure***
5. ***Preservation Rate**:*

   Preservation Rate = $|A \cap B| / |A|$

   A = Set of k-nearest neighbors in the original space

   B= Set of k-nearest neighbors in the reduced space

   $|A \cap B|$ *= Number of neighbors preserved after reduction*

6. ***Average Over All Data Points***

Notes:

- You are allowed to work in teams of up to 2.
- Submit a **notebook/file** with your code and a **report** detailing your work.
- Teams may lose credit in the discussion if any member has not contributed.
- Any cases of cheating will result in a score of 0% for all involved teams.